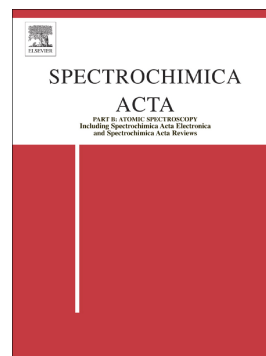


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Elemental and mineralogical imaging of a weathered limestone rock by double-pulse micro-Laser-Induced Breakdown Spectroscopy

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Abstract

The present work aims to evaluate the alteration conditions of historical limestone rocks exposed to urban environment using the Laser-Induced Breakdown Spectroscopy (LIBS) technique. The approach proposed is based on the microscale three dimensional (3D) compositional imaging of the sample through double-pulse micro-Laser-Induced Breakdown Spectroscopy (DP- μ LIBS) in conjunction with optical microscopy. DP- μ LIBS allows to perform a quick and detailed in-depth analysis of the composition of the weathered artifact by creating a ‘virtual thin section’ (VTS) of the sample which can estimate the extent of the alteration processes occurred at the limestone surface. The DP- μ LIBS analysis of these thin sections showed a reduction with depth of the elements (mainly Fe, Si and Na) originating from atmospheric dust, particulate deposition and the surrounding environment (due to the proximity of the sea), whereas, the LIBS signal of Ca increased in intensity from the black crust to the limestone underneath.

Keywords: Limestone, DP- μ LIBS, Microscope, Compositional imaging, 3D analysis

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1. Introduction

The outcrop of limestones as building material in the region of Apulia (southern Italy) has represented for centuries an important resource. In particular, limestone quarrying increased especially in the Middle Ages between the 12th and the 17th century AD, by Normans, Swabians and Angevins [1]. At the moment, Apulian limestones contribute to 10-15% of the Italian production of building stones [1], thanks to the number of aesthetical and technical features that support their use for a variety of architectural and ornamental applications.

However, after the rock is moved from its natural environment and put into operation, its degradation begins and continues on dependence on the new environment in which the building is located [2]. In particular, weather conditions play a relevant role in natural ageing of monuments, in addition to atmospheric pollution that affects the degradation of building stones due to chemical reactions caused by dry and wet depositions. Compact limestones mainly undergo superficial degradation by dissolution of carbonatic components, sulphatation processes and deposition of atmospheric particulate, which requires extensive, expensive and frequent cleaning works [3-6].

The composition, shape, and distribution of minerals in a rock depends mostly on its formation mechanisms and environment. Thus, the knowledge of the spatial distribution and chemical composition of minerals is of importance to understand the textural development and history of the material. Electron-probe microanalysis (EPMA) and micro-x-ray fluorescence (μ XRF) are generally used to analyse geological samples and obtain X-ray compositional mapping [7-8]. The EPMA features an image with analytical spatial resolution greater than μ XRF, especially when coupled with energy-dispersive spectrometry (EDS). However, samples to be analyzed by EPMA must be flat, polished and require carbon coating, thus mapping of large samples involve a number of sample preparation and acquisition issues.

During the last decades laser-induced breakdown spectroscopy (LIBS) has emerged as a very promising analytical technique in several application areas, including industrial, environmental and cultural heritage fields, and on materials in the solid, liquid or gaseous state [9,10]. Compared to the

analytical techniques mentioned above, LIBS features several advantages such as simplicity and robustness of the instrumentation, no or minimal sample preparation, and operation in ambient atmosphere. Due to the intrinsic speed and high spatial resolution of the technique, LIBS has been successfully used also for obtaining the compositional mapping of flat or more complex surfaces [11-17] and providing in-depth information by shooting several laser pulses on the same spot [18-22]. Further, coupling of LIBS with an optical microscope allows the improvement of the spatial resolution of the system, which is limited by the size of the laser-induced crater.

The present work aims to study the alteration conditions of an historical limestone sample exposed for decades to urban environment, using the double-pulse micro-Laser-Induced Breakdown Spectroscopy (DP- μ LIBS) approach. By imaging and identifying the mineralogical composition of the different layers, the technique can provide microscale three dimensional (3D) compositional information on the sample nature and degradation state avoiding the use of classical thin petrographic section microscope analysis and the long and complex sample preparation procedure involved.

2. Material and methods

The sample used in this work is a weathered limestone fragment collected from a quoin of the left jamb of the southern entrance gate to the courtyard of Castello Svevo, Bari (Italy). The castle, which features a quadrangular shape and square towers, is an historic “multilayered” monument built originally in 1131 by Roger the Norman on the remains of a Byzantine structure, and resulting of several building phases ranging from the 12th to the 16th century. This specific area of the masonry blocks of the limestone castle was chosen because it appears particularly degraded, with a deposit of black crusts accumulated on its surface over the centuries.

The analysis of the limestone sample was performed using a micro-LIBS setup based on the Modì-double-pulse instrument manufactured by Marwan Technology (Pisa). The μ -Modì double-pulse system worked with a double-pulse Nd:YAG laser by Lotis that emitted two collinear pulses at

1064 nm. The laser beams were directed through the Modì articulated arm to a Zeiss AxioPlan A1 optical microscope and then focused on the sample with a 10x magnification objective. To minimize the size of the laser-induced crater, while maintaining the benefits of double-pulse LIBS analysis, an asymmetric configuration was used with the first laser pulse energy of 5.4 mJ and the second laser pulse energy of 8.7 mJ [23]. The full width at half-maximum (FWHM) of the laser pulses was 20 ns and the delay between the pulses 1 μ s. In these conditions, the diameter of the craters produced on the limestone sample was about 20 μ m, with a depth of about 1 μ m. The LIBS signal was collected by an optical fiber placed at 45° with respect to the laser direction, at a distance of about 1 cm from the sample, using a ball lens in front of the fiber that ensured the optimal collection of the signal from the whole plasma. The LIBS signal was then analysed using a double spectrometer (AvaSpec-USB2.0 by Avantes) covering the spectral range from 190 to 900 nm with a resolution of 0.1 nm in the UV region (190 to 450 nm) and 0.3 nm in the VIS-IR region (450 to 900 nm). The minimum acquisition time of the spectrometer was about 2 ms, i.e. much longer than the plasma lifetime, thus the spectral acquisition was not time-resolved.

Further, the microscope plate used was equipped with a two axis pc-controlled motor by ThorLabs to ensure the movement of the sample stage. A homemade software (LabWiev 8.5) allowed to synchronize the lasers, motors and spectrometer. The in-depth analysis was realized by delivering multiple laser pulses on the same point of the sample.

The elemental imaging was performed on the sample rough surface without further preparation, using a grid of analytical spots of about 20 μ m diameter, spaced at 200 μ m intervals, on a 5-mm² square surface (625 points) and for 5 (double) laser pulses in order to evaluate the change of elemental composition with depth. A spectrum was acquired for each analyzed point (3125 spectra in total). The time needed to acquire the 3D map was around 35 min.

The 3D compositional map was generated plotting the intensity of the LIBS emission line of the elements of interest at the pixel (x,y) corresponding to the x and y steps of the raster (200 μ m), and at a z coordinate representing the depth of the laser crater, which was estimated by microscopic

analysis of the samples to be about 1 μm per laser pulse. Table 1 lists some of the elements analysed and the wavelengths of the peaks used to construct the elemental maps. The spectral lines reported were chosen among the most intense of the corresponding elements, trying to avoid lines that could be affected by interferences from other neighboring lines.

Table 1. Peak wavelengths of the elements considered.

<i>Element</i>	<i>Wavelength (nm)</i>
<i>C I</i>	247.86
<i>Mg I</i>	285.21
<i>Al I</i>	309.27
<i>Si</i>	288.16
<i>Ti II</i>	337.28
<i>Fe II</i>	259.94
<i>Ca I</i>	422.68
<i>Sr II</i>	407.77
<i>Na I</i>	588.99 and 589.59
<i>K I</i>	766.49

No attempt was made to obtain a quantitative estimation of the elements concentrations, as the analysis performed aimed to the estimation of the in-depth and surface compositional changes produced on the sample by environmental factors. Most of the lines chosen for the 3D elemental imaging are resonant. Although the use of resonant lines would be inappropriate for quantitative analysis, the possible effects of self-absorption of these lines are not expected to prevent the study of the qualitative changes of line intensity with depth, and the build of 3D compositional maps and virtual thin sections (VTS).

3. Results and discussion

The limestone fragment (Fig. 1) showed an evident superficial deterioration with the presence of black crust. Averaging the intensity values of the elements of interest at different depths under the surface provided information on the variation of the element concentration inside the black crust, which allowed to estimate its depth and nature. In particular, marked differences of elemental composition were observed among surface-averaged LIBS emission intensities (Fig. 2) with increasing depth from laser pulse 1 (the superficial one) to laser pulse 5 (the deepest one).

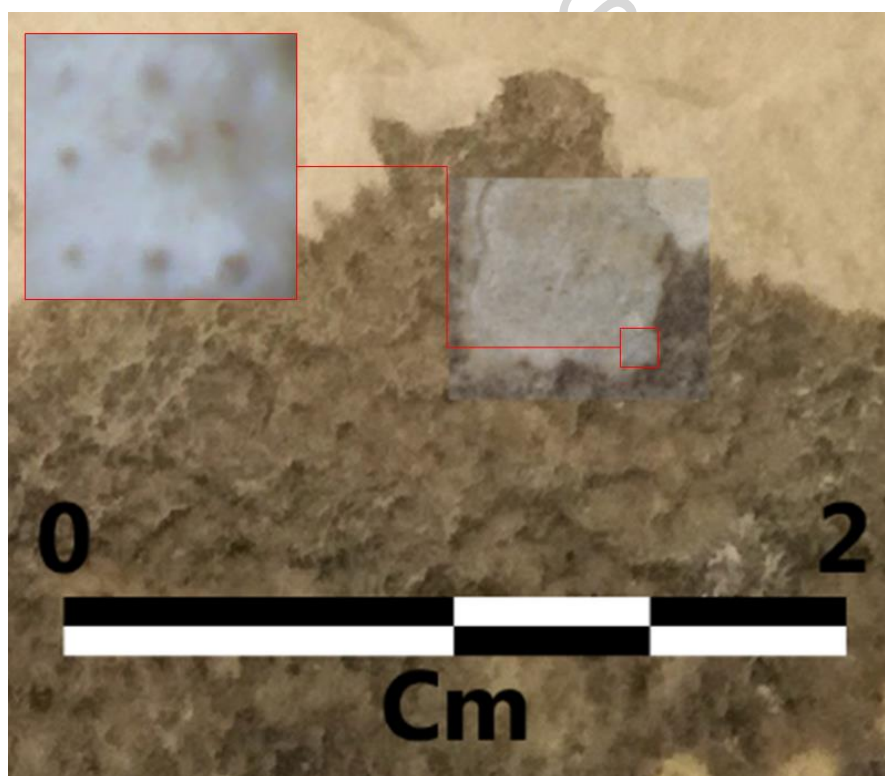


Fig. 1. The limestone fragment analysed using DP- μ LIBS 3D imaging. The area analyzed is that included in the square. In the inset, a detail of the area showing the laser craters.

Fig. 2 shows the variations (relative to the surface) of the line intensities of the elements of interest with depth (down to about 5 μ m under the sample surface). The line intensities were normalized to the average total spectral intensity at each layer. The indetermination on the surface-averaged line

intensities is not shown in Fig. 2 for the sake of readability. However, the corresponding standard deviations were lower than 3% for all the lines considered.

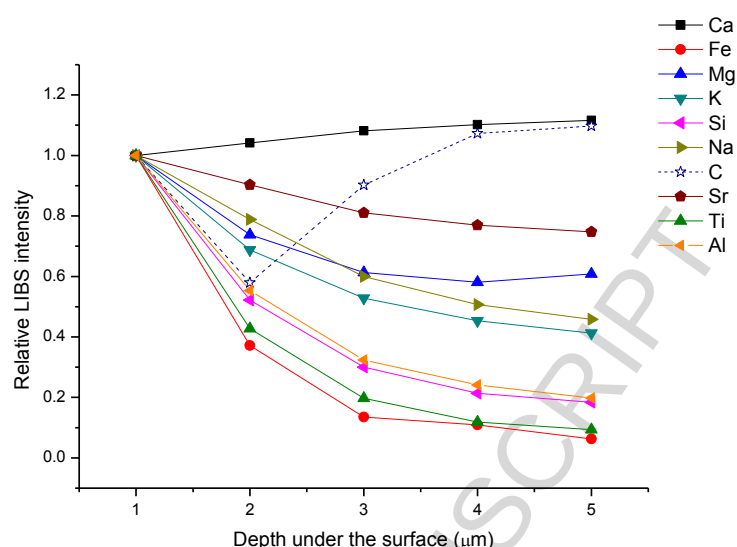
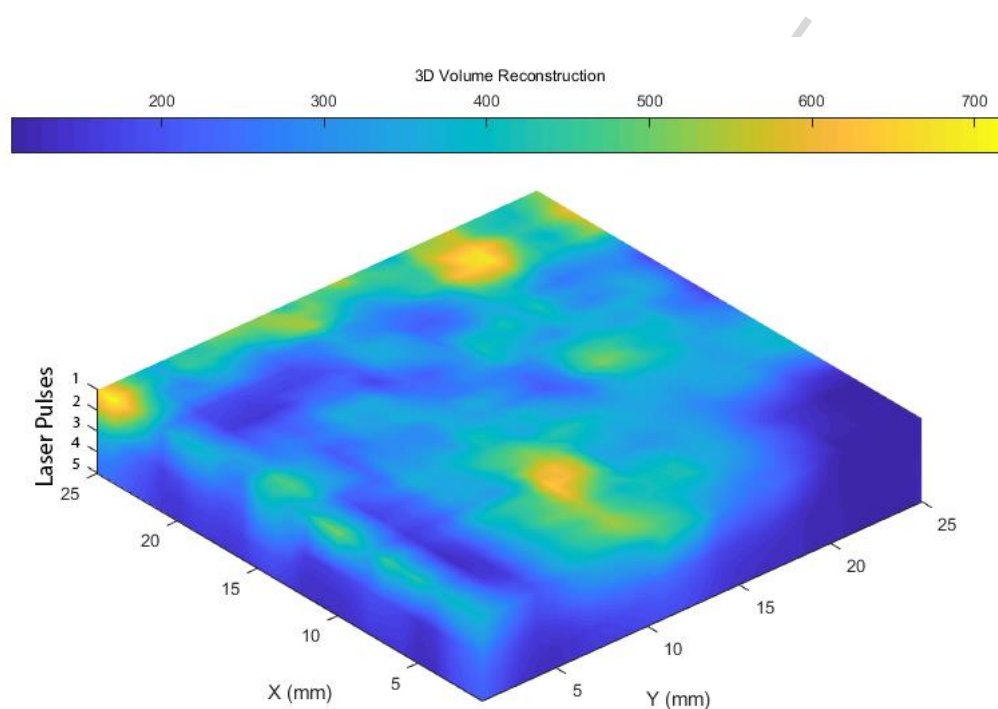


Fig. 2. Surface-averaged LIBS intensities of the elements considered at 5 laser pulses in depth.

Trends in Fig. 2 show that the elemental concentrations reached an almost constant value between the fourth and the fifth laser (double) pulse (4-5 μm under the surface). Thus, the thickness of the black crust in the specific point analysed can be estimated to be approximately 4-5 μm . Further, the concentration of the main elements in the black crust decreased with increasing depth, except C and Ca, which are the major component of the limestone.

The presence of Al, Fe, Si and Ti in the black crust may be ascribed to atmospheric pollution and dust and particulate deposition, whereas the presence of Na may be due to the marine aerosol that pervades the urban area in which the monument is located. The decrease of Sr concentration with depth would suggest a possible environmental source. The concentration of C initially decreases, as for the other elements originating from the environment, then increases again up to reach its surface value very similar to that of Ca. The LIBS signal of Mg also decreases with depth with a slight increase in the last laser pulse/layer at the interface between the black crust and the unweathered limestone underneath, which suggests a possible dedolomitization process of the rock [24].

The above considerations are based on the information obtained from the surface-averaged maps of the LIBS spectra, whereas the full 3D elemental maps obtained by DP- μ LIBS (Fig. 3, note that the z-axis is not in scale with the x and y axes) allow a more detailed analysis of the composition of the weathered limestone examined.



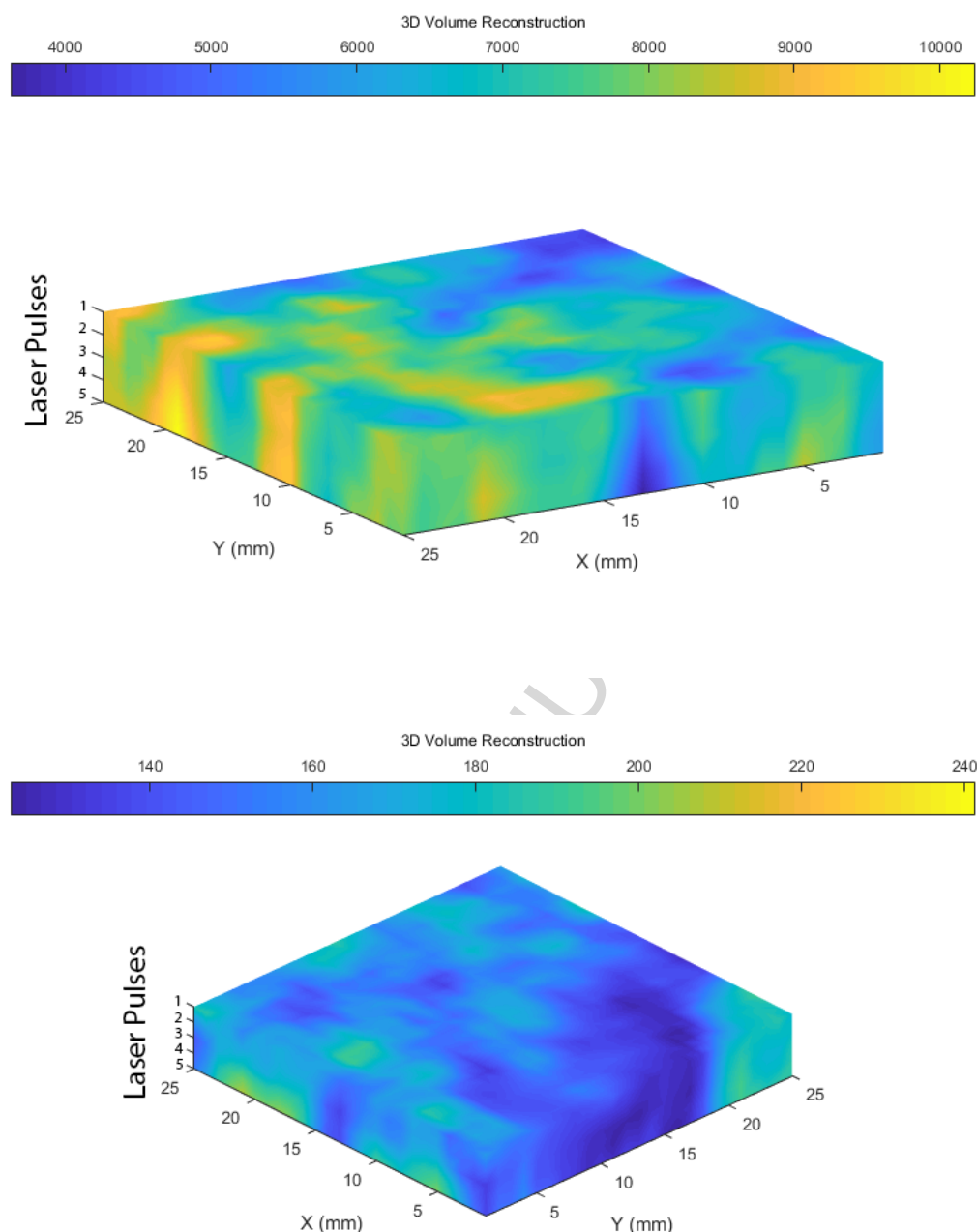


Fig. 3. 3D maps of Fe (top), Ca (middle) and C (bottom).

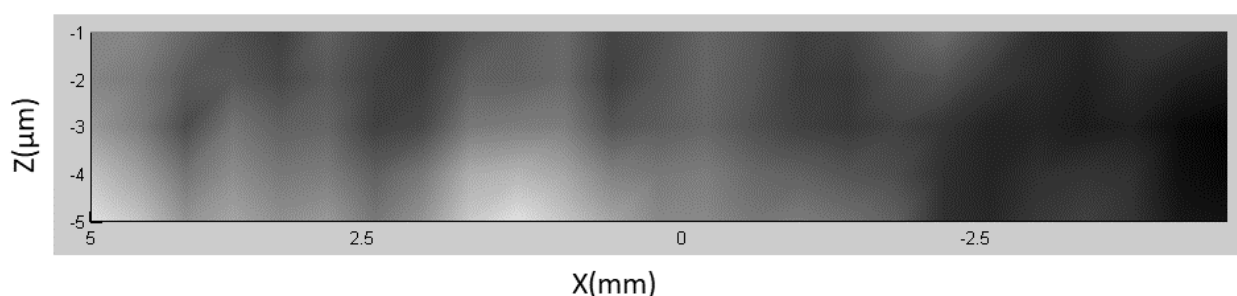
Further, videos of the reconstructed 3D volumes of Fe, Ca, and C were obtained using the Alan Jennings function (Copyright © 2013, Alan Jennings, All rights reserved), and are shown in Fig. S1 as supplementary material. The 3D maps in Fig. 3 confirm the general trends shown in Fig. 2, i.e. the LIBS signal of Fe decreases with depth in the black crust, that of Ca increases, and the behaviour of C is more complex but suggests an initial decrease followed by an increase close to the limestone surface. In any case, the composition of the black crust appears quite inhomogeneous.

Further, similar results (not shown here) were obtained by measuring and imaging by the same technique another clean limestone sample and a degraded portion of it before and after crust removal by five laser double pulses.

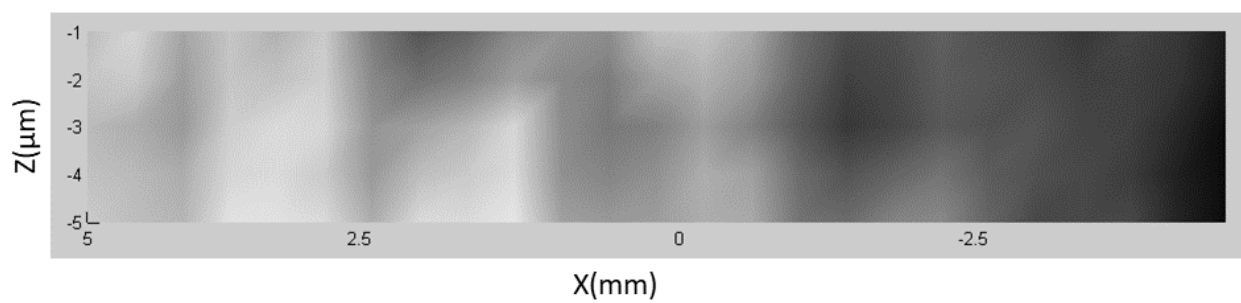
From a mineralogical point a view, it would be interesting to infer the association of different elements in the volume. For example, the similar trend shown by Al and Si concentrations with depth (Fig. 2) would suggest their presence in the black crust in the form of alumino-silicates. The surface of a rock may also include minerals that originate from partial interactions of the rock surface with its environment over a certain period of time. In principle, 3D maps are a powerful tool to show differences from weathering phenomena and geological processes involved in rock transformations but their interpretation can be difficult for geologists not accustomed to this kind of visualization.

A ‘virtual thin section’ (VTS) of the sample obtained by slicing the 3D compositional maps perpendicularly to the sample surface would provide an information more easily interpretable. The elemental sections shown (in grey scale) in Fig. 4 were obtained ‘cutting’ the sample arbitrarily by a plane perpendicular to the surface at $y=0$.

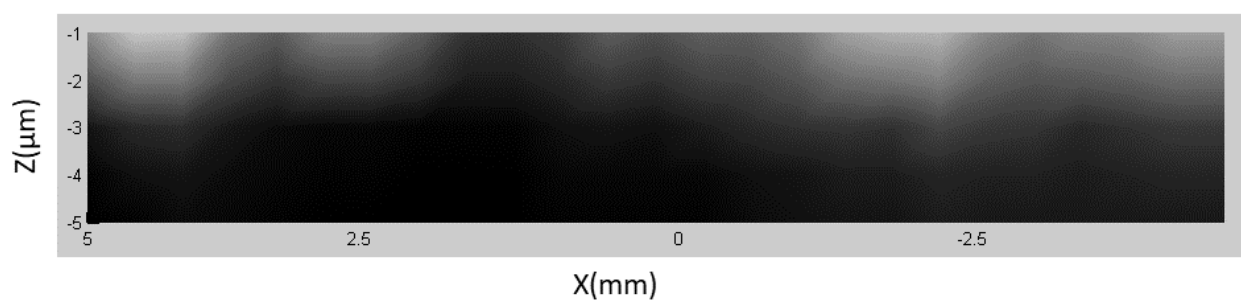
C



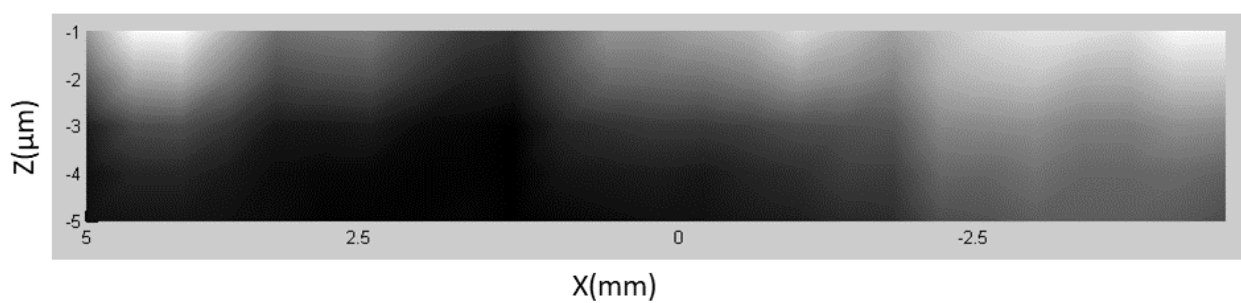
Ca



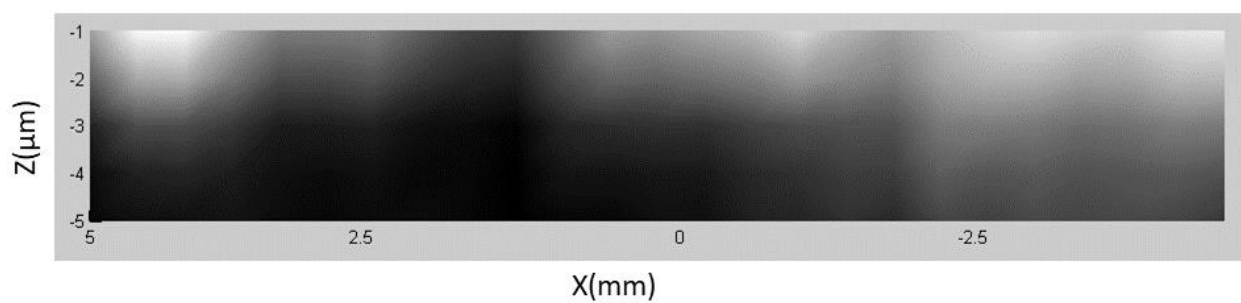
Fe



Al



Si



Mg

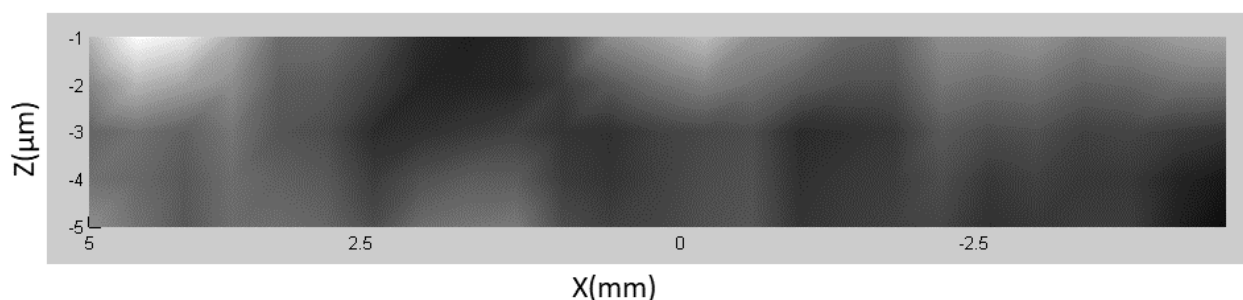


Fig. 4. Elemental ‘virtual thin sections’ (VTS) of the main elements present in the black crust.

The grey scale elemental ‘sections’ can be combined to form three-chromatic false-color images. In particular, false color red-green-blue (RGB) maps were constructed assigning the intensity map of Al to the red channel, of C to the green channel and Si to the blue channel, whereas the magenta color (Red+Blue) indicates the co-presence of Al and Si (alumino-silicates) (Fig. 5). Similarly, a three-chromatic image with Al in the red channel, Mg in the green and Fe in the blue was constructed (Fig. 6), and so on. The yellow zones in Fig. 6 correspond to high simultaneous concentrations of Red+Green, i.e. Al+Mg, magenta zones corresponds to Red+Blue, simultaneous high concentrations of Al and Fe, while the cyan color (Green + Blue) indicates high simultaneous concentrations of Mg and Fe. Finally, the white zones correspond to the co-presence, at high concentrations, of all the elements considered in the map. Although this kind of imaging may help in identifying the compositional variation in the sample, the determination of the mineralogical nature of the black crust remains a difficult task to perform because only three elements can be mapped in a single section, as the number of bands that can be visualized in a color image is limited to three (RGB).

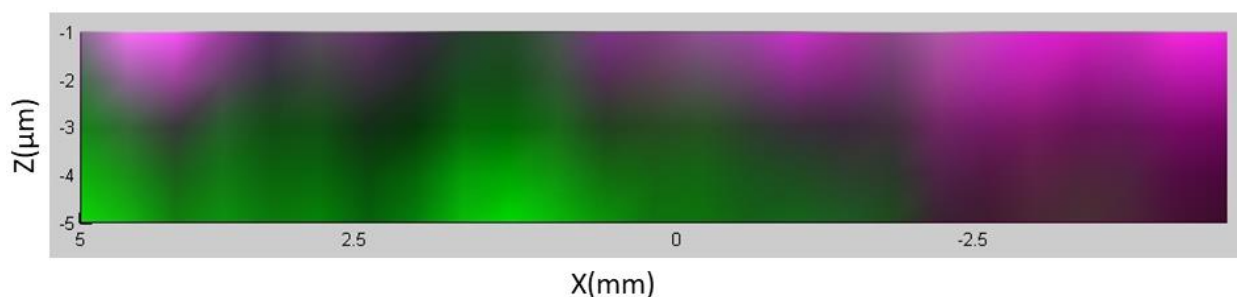


Fig. 5. ‘Virtual thin section’ of the black crust, evidencing the presence of alumino-silicates (magenta regions).

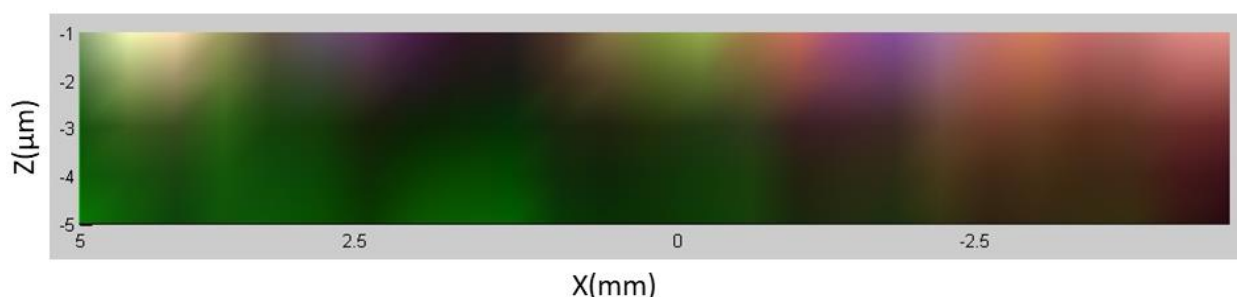


Fig. 6. ‘Virtual thin section’ of the black crust (Red=Al, Mg=Green, Fe=Blue).

For a quick evaluation and analysis of the ‘virtual thin sections’ the method introduced by Pagnotta et al. [20] was used, which allowed to obtain information about the mineralogical structure of the black crust by the simultaneous elaboration of the compositional maps of the elements considered in the analysis. This approach, based on the Self-Organized Maps (SOM) method developed by Kohonen [25], considers the contribution of all the elements in the ‘virtual thin section’ to construct a single mineralogical map. The SOM method is an unsupervised classification method which does not perform any dimensional reduction, thus it appears particularly suited for clustering in multi-dimensional spaces, as the one defined by the LIBS signal of the elements considered (Al, C, Ca, Fe, Mg, Si). The results obtained (Fig. 7) show the change of mineralogical composition in the transition from the black crust layer to the limestone underneath. The average composition of the clusters provided by the SOM algorithm is shown in Fig. 8. The two first clusters, represented,

respectively, in red and green in Fig. 7 have similar composition (C, Ca and Mg) with a higher percentage of Al and Fe in the upper red zone with respect to the lower green part. The blue zone is dominated by Al, Si and C, while the black part is mainly composed by Al and Si (see also Fig. 5).

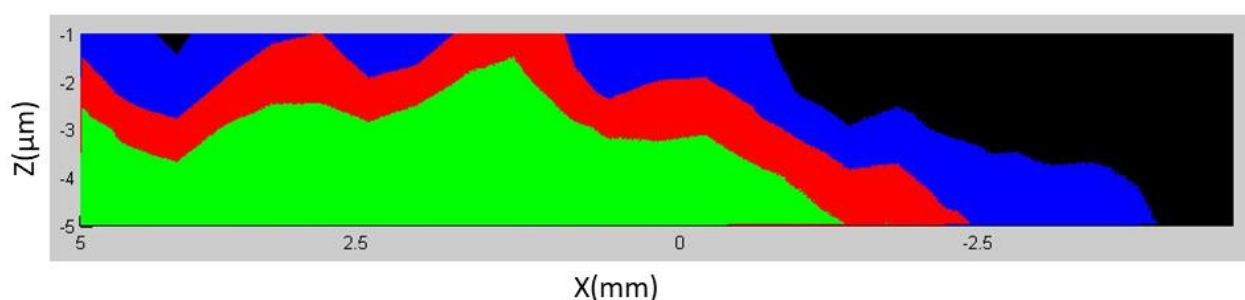


Fig. 7. False-color representation of the 'virtual thin section' of the black crust according to the SOM method.

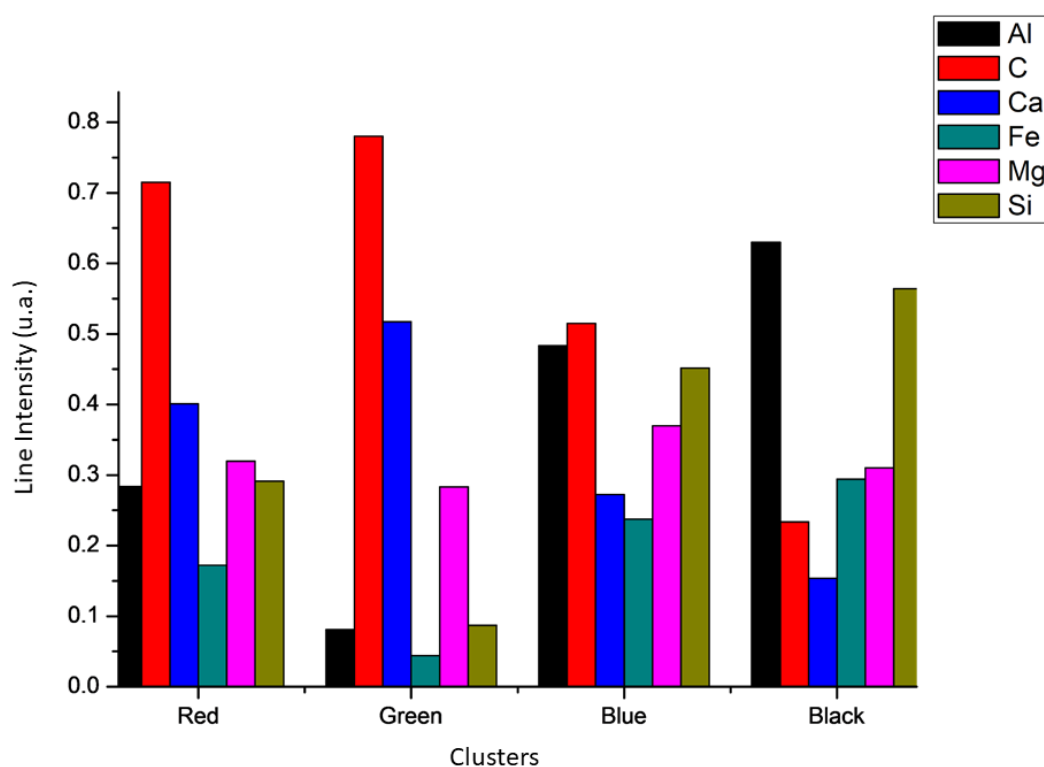


Fig. 8. Average composition of the clusters in Fig. 7.

4. Conclusions

The DP- μ LIBS technique appeared to be very effective for obtaining a high-resolution and fast 3D compositional imaging of the degradation layer (black crust) on the limestone rock studied. The decrease of Al, Fe, Si and Ti line intensities and the increase of Ca concentration with depth when passing from the superficial black crust layer to the rock underneath, could be ascribed to the decreased atmospheric pollution effects. The high concentration of Na in the superficial layers could be ascribed mainly to external sources such as marine aerosol.

In conclusion, the DP- μ LIBS technique allows the analysis of the degradation layer of historical limestones without any treatment by the construction of 'virtual thin sections' that can provide on the elemental and mineralogical composition of the sample. This technique is able also to detect elements of low atomic mass, i.e. C, and does not suffer from limitations involved with other more conventional techniques, such as EPMA, μ XRF, LA-ICP-MS. Besides its capability of assessing the variation of specific elemental and mineralogical components of a weathered limestone rock with depth, the technique appears promising also for investigating rock samples of different origin and nature and their alteration processes due to environmental factors.

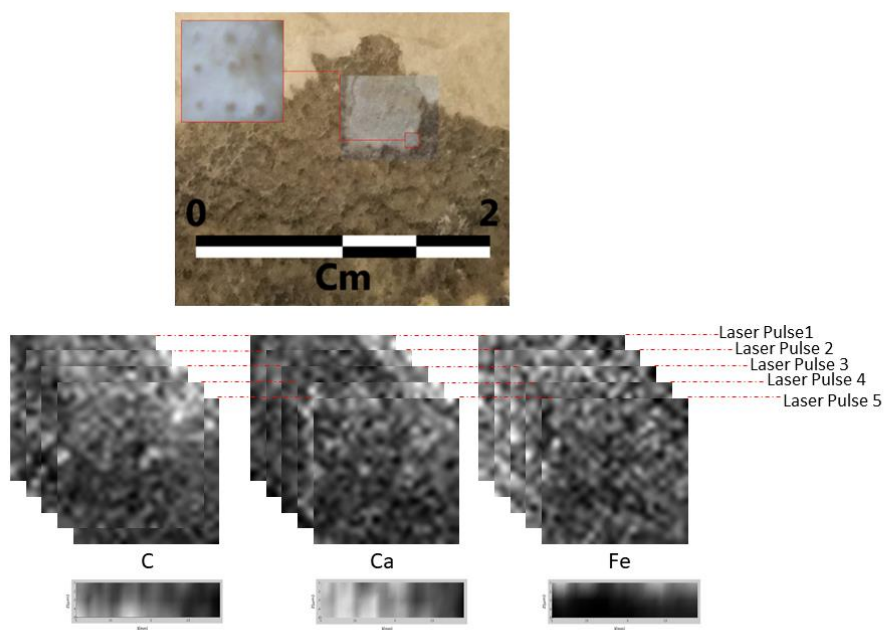
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Graphical abstract



Highlights

- High-resolution and compositional imaging for the evaluation of the alteration of historical rock;
- 3D compositional imaging of the weathered limestone through DP- μ LIBS;
- DP- μ LIBS allows to perform in-depth analysis of the composition of the weathered rock;
- ‘Virtual thin section’ can estimate the alteration processes occurred at the limestone surface.

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